# Pre-work · Intro to Anatomy

Intro to Anatomy. Read the Part A chapters first, then work this in order. You come to class ready to use this, not to hear it for the first time.

Module 1 · 1 of 4 | Bring this to class

## How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

**Why this order.** Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

## Organize it

Turn the two intro chapters into mini visuals. Seven chunks, each in the shape that fits it.

### FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

### Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

## Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

### HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

## Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

### WORKED EXAMPLE

**The elbow is ______________ to the wrist.**

**Answer.** Proximal.

**Reasoning.** Both structures are on the same limb, so the proximal and distal pair is available. Proximal means nearer the point of attachment to the trunk. The elbow attaches nearer the shoulder than the wrist does, so the elbow is proximal. Note the wrong answer you could reach for: the elbow is also superior to the wrist in anatomical position, but that is the weaker answer, because the question is about a limb and the limb-specific pair is the precise one.

### Set A · Directional relationships

1. The sternum is ______________ to the vertebral column.
2. The lungs are ______________ to the diaphragm.

takes longer than three minutes you are copying instead of organizing.

1 **LADDER**

### The six levels of structural organization

Six rungs, simplest at the bottom. Write one real example beside each rung, not the one from the notes.

2 **SPLIT**

### The seven directional pairs

One split per pair. Pair on the branches, and underneath, the one sentence that proves you can use it. Star the two you are least sure of.

3 **GRID**

### Planes crossed with what they divide

Planes down the side, the two parts each produces across the top. Six rows. Draw the cut line on a stick figure in one box.

4 **HUB**

### Body cavities

Ventral and dorsal as two hubs. Spoke out to each subdivision, then to what it contains. Mark the diaphragm on the spoke it actually sits on.

5 **CHAIN**

### Serous membrane, wall to organ

Body wall, arrow, parietal layer, arrow, cavity with fluid, arrow, visceral layer, arrow, organ. Then write the chain three times, once each for pericardium, pleura, peritoneum.

6 **HUB**

### The mediastinum

Borders as spokes on one side, contents as spokes on the other. Five borders, five contents.

**DRAWING 1**

### Anatomical position, with the planes on it

**BRAIN DUMP FIRST**

Two minutes. Anatomical position and every plane you can name, with what each one separates.

Draw a full figure in anatomical position. Mark the five features that define it: feet flat, arms at the sides, palms forward, head and eyes forward, body erect. Then cut the same figure with the sagittal, frontal, and transverse planes. Label each, and write the two portions each one creates.

In your notebook, head the page **M1–1 Anatomical position, with the planes on it** so you can find it later. Correct it in a second colour.

**YOU KNOW IT WHEN**

You can use midsagittal and parasagittal on one figure and say the difference in a sentence.

3. The thumb is ______________ to the little finger, in anatomical position.
4. The skin of the thigh is ______________ to the femur.
5. A stroke in the left cerebral hemisphere causes weakness on the right side of the body. That effect is ______________________.
6. Explain in one sentence why it is wrong to say the nose is proximal to the chin.

### Set B · Planes and sections

1. A scan of the head shows both eyes and both ears in the same image, with the top of the skull missing. Which plane produced it, and how do you know?
2. A cut separates the body into anterior and posterior portions. Name the plane.
3. A section through the arm shows a single round bone surrounded by a ring of muscle. Which plane, and how do you know?
4. You want an image showing the relationship between the front of the heart and the sternum. Which plane serves you best, and why?

### Set C · Cavities and membranes

1. A stab wound opens the pleural cavity. Predict what happens to the lung on that side, and say why.
2. A patient has a ruptured appendix. Name the membrane involved, the condition it produces, and why the problem does not stay local.

7 **GRID**

### Quadrants and regions on one abdomen

Draw the abdomen once. Quadrant lines in one colour, the nine-region lines in a second. Label the organs where they sit. One drawing does both systems.

**YOU KNOW PANEL 1 WORKED WHEN**

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

**DRAWING 2**

### The cavities, lateral view

**BRAIN DUMP FIRST**

Two minutes. Every cavity, what is in it, and what separates it from the one next door.

Draw the body in profile. Outline the dorsal cavity and the ventral cavity, then subdivide each. Shade the thoracic and abdominopelvic cavities differently and draw the diaphragm between them. Put the mediastinum in, and shade the retroperitoneal space behind the peritoneum.

In your notebook, head the page **M1–2 The cavities, lateral view** so you can find it later. Correct it in a second colour.

**YOU KNOW IT WHEN**

You can point at any organ on your drawing and say which cavity it sits in and which membrane wraps it.

3. A surgeon plans to reach the right kidney. Is the approach through the peritoneal cavity or behind it? Name the term and one other structure in the same space.

4. A friction rub is heard over the heart. Name the two layers rubbing, the fluid that should be preventing it, and the condition.

### Set D · Locate it

1. A patient reports pain in the right lower quadrant. Name two organs that could be responsible.

2. Give the regional term for the front of the elbow, the back of the knee, and the sole of the foot.

3. Sort as axial or appendicular: brachial, thoracic, popliteal, vertebral, carpal.

**YOU KNOW PANEL 3 WORKED WHEN**

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

### Serous membrane, the fist and the balloon

BRAIN DUMP FIRST

One minute. Parietal, visceral, cavity, fluid. What each one touches.

Push a fist into a partly inflated balloon and draw that. The layer against your fist is visceral, the outer layer is parietal, the space between holds the fluid. Label all four. Then draw what happens to the space when it fills with air, and when it fills with fluid.

In your notebook, head the page **M1-3 Serous membrane, the fist and the balloon** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain a collapsed lung and a friction rub using only this drawing.

## Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

## 1 The six levels of structural organization

LADDER

## 2 The seven directional pairs

SPLIT

## 3 Planes crossed with what they divide

GRID

## 4 Body cavities

HUB

## 5 Serous membrane, wall to organ

CHAIN

### 6 The mediastinum

HUB

### 7 Quadrants and regions on one abdomen

GRID